

# A consistent dielectric response model for water ice over the whole energy–momentum plane

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## Abstract

We have developed, for the first time, a consistent analytic model for the complex dielectric response function,  $\epsilon$ , of amorphous and hexagonal ice over the complete energy–momentum plane. The energy dependence at the optical limit is based on a parametrization of all dielectric data representations, namely  $\text{Im}(\epsilon)$ ,  $\text{Re}(\epsilon)$  and  $\text{Im}(-1/\epsilon)$ , using a linear combination of Drude-type functions associated with the various interband transitions. The “fitting” procedure is forced to self-consistency by the fulfillment (to within 1%) of the  $f$ -sum-rules for  $\text{Im}(\epsilon)$  and  $\text{Im}(-1/\epsilon)$ . The momentum dependence is provided by appropriate dispersion relationships for the Drude coefficients obtained from our recent analysis of the experimental Bethe ridge of liquid water. Time-dependent density functional theory (TDDFT) calculations of the optical oscillator strength (up to 20 eV) of H<sub>2</sub>O clusters are also presented for comparison. It is expected that the present development will permit accurate calculations of inelastic scattering probabilities and associated quantities (e.g. stopping forces) of swift charges in solid water.

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## 1. Introduction

Over a broad range of impact energies inelastic cross sections and associated quantities (e.g. stopping forces) of swift charges in condensed media may be calculated “exactly” if the material’s complex dielectric response function  $\epsilon(E, q) = \epsilon_1(E, q) + i\epsilon_2(E, q)$  is known over the whole range of energy ( $E$ ) and momentum ( $q$ ) transfer [1]. To that end, it is important to recognize that although the dielectric function is formally defined through the many electron wavefunctions – which are difficult to obtain for realistic materials – it is also directly associated to various measur-

able properties (e.g. optical constants) which may be obtained experimentally [2]. Water ice in its amorphous and hexagonal forms is of particular interest since the former is perhaps most closely related to the liquid phase whereas the latter is the crystal structure encountered on Earth. Several dielectric models pertaining to water in the condensed-phase have been proposed over the last 30 years and reviewed elsewhere [3,4]. Almost all of them are for liquid water with only few exceptions for solid water (i.e. ice). Apart from the work of Long et al. [5], Zaider and co-workers [6, and reference therein] and, more recently, Hahn et al. [7] and Tavernelli [8], which adopt an *ab initio* approach (to various levels of sophistication), all other studies take direct advantage of the experimental data at the optical limit ( $q = 0$ ) already available since the early ’70s [9–11]. Thus, the construction of a dielectric model

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normally takes place in two steps; first, the optical data are represented by suitable analytic functions that have to fulfil certain theoretical constraints (e.g. sum-rules), and, second, the dielectric functions so obtained are extended to finite momentum ( $q > 0$ ) by physically motivated and computationally simple dispersion schemes [12]. The two perhaps most important aspects in this approach are, first, the use of experimental data at the optical limit which automatically accounts, for the most part, for many-body effects (e.g. polarization, correlation, collective excitations) which are computationally intractable for realistic materials and, second, the flexibility and generality of the method which can be essentially applied to any system, being an atom, molecule or condensed target, as long as the necessary optical data are available. However, aside any uncertainty in the optical data, insofar as the dispersion models used are untested, the level of uncertainty in the subsequent calculation of “exact” quantities (i.e. cross sections) remains, largely, also unknown.

For water (or other biomaterials) dispersion models of either the  $\delta$ -oscillator [13,14] or the extended-Drude [15] form have been used. Although in all cases the models lead to the correct behaviour at both the optical ( $q = 0$ ) and free-electron ( $q = \infty$ ) limits, we have recently shown that their agreement with the available experimental data for liquid water at intermediate values of  $q$  is unsatisfactory [16,17].

Along the lines of our work on liquid water [16], we have developed an analytic dielectric model for hexagonal and amorphous ice based on the extended-Drude scheme, which is consistent with the available optical data for the two materials and satisfy the  $f$ -sum-rules to within 1%. The momentum dependence is provided by our recently developed dispersion relationships for the Drude coefficients which are consistent with the experimentally obtained Bethe ridge of liquid water. The resulting characteristics of our model energy-loss-functions (ELFs) for the two forms of ice are presented.

## 2. Methodology

The optical limit ( $q = 0$ ) of our dielectric model has been obtained from an appropriate linear combination of Drude-type functions representing the various interband transitions of ice. The appropriate modification and generalization of the Drude free-electron model for the conduction band of metals to a system of damped linear oscillations corresponding to the more complex band structures of real solids has been well established [18], and used widely with considerable success [12] largely due to its computational simplicity and analytic properties (sum-rules, Kramers-Kronig pairs etc.). The Drude model parameters are adjusted against all dielectric data representations, namely,  $\text{Im}(\epsilon)$ ,  $\text{Re}(\epsilon)$  and  $\text{Im}(-1/\epsilon)$ , obtained from EELS [19] and reflectance [20] measurements on amorphous and hexagonal ice, respectively. The experimental data for both forms of ice extend from threshold up to  $\sim 30$  eV covering the most important part of the

valence spectral range. Since the  $\text{Im}(\epsilon)$  is proportional to the optical oscillator strength, its partition to the various interband transitions has been guided by the properties of the absorption spectrum of the  $\text{H}_2\text{O}$  molecule. However, the latter is allowed to vary within reasonable limits to reflect the influence of the long-range intermolecular forces which show up in the condensed phase [21,22]. Thus, the spectral distribution of  $\text{Im}(\epsilon)$  is obtained from a superposition of the four-outer valence shells of  $\text{H}_2\text{O}$  with “fixed” ionization energies ( $B_i = 10, 13, 17, 32$  eV) and of five-discrete excitation levels with characteristic energies that are distributed “freely” between 8 and 15 eV (band-gap:  $B_{\min} \approx 7\text{--}8$  eV). These values have been selected as a reasonable compromise between the various theoretical and experimental data available for the various condensed forms of water [21,22]. In accordance with both theoretical and experimental findings no account of a plasmon excitation is considered [23,24]. Similar to our liquid model [16] and that of others [25], we have found convenient to represent each ionization channel by a (normal) Drude function and each excitation/discrete channel by a derivative-type Drude function. This distinction is made on purely practical grounds since the derivative function is more sharply peaked and, thus, facilitates the fit of the data at the low energy part of the spectrum. The parametrization of the experimental data for the  $\text{Im}(\epsilon) = \epsilon_2$  of the valence shells ( $v$ ) is then obtained by the following set of equations:

$$\epsilon_2^{(v)}(E, q \approx 0)_{\text{exp}} = \begin{cases} E_p^2 \left[ \sum_{i=1}^4 D_2(E; f_i, E_i, \gamma_i) + \sum_{j=1}^5 \tilde{D}_2(E; f_j, E_j, \gamma_j) \right] & E > B_i \\ E_p^2 \sum_{j=1}^5 \tilde{D}_2(E; f_j, E_j, \gamma_j) & B_{\min} \leq E < B_{i,\min} \end{cases} \quad (1)$$

where

$$D_2(E; f_i, E_i, \gamma_i) = \frac{f_i \gamma_i E}{(E_i^2 - E^2)^2 + (\gamma_i E)^2}, \quad (2a)$$

$$\tilde{D}_2(E; f_j, E_j, \gamma_j) = \frac{2f_j \gamma_j^3 E^3}{\left[ (E_j^2 - E^2)^2 + (\gamma_j E)^2 \right]^2}. \quad (2b)$$

The symbols are as follows:  $D$  and  $\tilde{D}$  represent the standard (or normal) and derivative forms of the Drude function, respectively;  $E_p = 4(\pi n \alpha_0^3)^{1/2} R$  is the nominal free-electron plasma energy of the material with electronic density  $n$ ,  $\alpha_0$  is the Bohr radius and  $R$  is the Rydberg constant ( $E_p$  is 20.82 and 20.59 eV for the amorphous and hexagonal forms of ice, respectively); the triads  $\{f_{i,j}, E_{i,j}, \gamma_{i,j}\}$  represent the apparent oscillator strengths, damping energies, and transition energies, respectively, of the various inelastic channels considered in the deconvolution of the experimental optical spectra for  $\text{Im}(\epsilon)$ . These coefficients are practically associated with the height, width and position of each peak that contributes to the absorption spectrum and were, therefore, used as adjustable model parameters. Thus, the procedure starts with a first guess of values obtained from a fit to the experimental data

for  $\text{Im}(\varepsilon)$  by Eqs. (1) and (2) and these values are then used to fit the data for  $\text{Re}(\varepsilon)$  using the following expressions:

$$\varepsilon_1^{(v)}(E, q \approx 0)_{\text{exp}} = 1 + E_p^2 \left[ \sum_{i=1}^4 D_1(E; f_i, E_i, \gamma_i) + \sum_{j=1}^5 \tilde{D}_1(E; f_j, B_j, \gamma_j) \right], \quad (3)$$

where

$$D_1(E; f_i, E_i, \gamma_i) = \frac{f_i(E_i^2 - E^2)}{(E_i^2 - E^2)^2 + (\gamma_i E)^2}, \quad (4a)$$

$$\tilde{D}_1(E; f_j, E_j, \gamma_j) = \frac{f_j(E_j^2 - E^2) [(E_j^2 - E^2)^2 + 3(\gamma_j E)^2]}{\left[ (E_j^2 - E^2)^2 + (\gamma_j E)^2 \right]^2}. \quad (4b)$$

A new set of values is likely to be obtained after the application of Eqs. (3) and (4) which are further tested against the experimental data for  $\text{Im}(-1/\varepsilon)$  using the expression:

$$\text{Im} \left[ \frac{-1}{\varepsilon^{(v)}(E, q=0)} \right] = \sum_{i,j}^{\text{valence}} \frac{\varepsilon_2^{(i,j)}(E, 0)}{|\varepsilon^{(v)}(E, 0)|^2}. \quad (5)$$

The procedure is repeated until all dielectric data representations are reproduced satisfactorily by the model functions. The above “fitting” procedure is forced to self-consistency by the simultaneous fulfillment (to within 1%) of the  $f$ -sum-rules for  $\text{Im}(\varepsilon)$  and  $\text{Im}(-1/\varepsilon)$  which in the normalized form may be written as:

$$N_{\text{eff}}^{(v)} = \frac{2}{\pi E_p^2} \int_0^\infty E \text{Im}[\varepsilon^{(v)}(E, 0)] dE = 1 - N_{\text{eff}}^{(K)}, \quad (6a)$$

$$N_{\text{eff}}^{(v)} = \frac{2}{\pi E_p^2} \int_0^\infty E \text{Im} \left[ -\frac{1}{\varepsilon^{(v)}(E, 0)} \right] dE = 1 - N_{\text{eff}}^{(K)}, \quad (6b)$$

where  $N_{\text{eff}}^{(K)} = 0.178$  [26]. Following the standard practice it is assumed that the inner K-shell which sets in at 532 eV conserves its atomic character, i.e.  $\text{Re}(\varepsilon^{(K)}) \approx 1$ . Thus, the K-shell contribution to the total  $\text{Im}(\varepsilon)$  and  $\text{Im}(-1/\varepsilon)$  was obtained from its optical-oscillator-strength calculated from the photoionization data of NIST [27]. The latter are converted to  $\text{Im}(\varepsilon^{(K)})$  and fitted by a single (normal) Drude function. Then, the total optical-ELF is obtained from:

$$\text{Im} \left[ \frac{-1}{\varepsilon(E, q=0)} \right] \approx \sum_{i,j}^{\text{valence}} \frac{\varepsilon_2^{(i,j)}(E, 0)}{|\varepsilon^{(v)}(E, 0)|^2} + \varepsilon_2^{(K)}(E, 0). \quad (7)$$

A consistency test for the optical-ELF obtained from Eq. (7) is the calculation of the mean excitation energy of the stopping power theory, the so-called  $I$ -value defined by:

$$\ln(I) = \frac{\int_0^\infty E \ln(E) \text{Im} \left[ \frac{-1}{\varepsilon(E, 0)} \right] dE}{\int_0^\infty E \text{Im} \left[ \frac{-1}{\varepsilon(E, 0)} \right] dE}. \quad (8)$$

Obviously, in the absence of a realistic band structure model for the materials examined the partition of the optical spectrum to the various interband transitions is tentative. Nevertheless, the above procedure being consistent with all dielectric data representations as well as with the theoretical constraints is expected to minimize the uncertainties involved in the procedure. The Drude model parameters obtained in the present study for the two forms of ice examined are provided in Table 1.

The  $q$ -dependence in our dielectric model has been deduced from an analysis of the experimental data on the Bethe ridge of liquid water as discussed elsewhere [16]. Along the lines of the extended-Drude scheme we extend our dielectric model to  $q > 0$  by appropriate dispersion relationships for the Drude coefficients as following:

$$f_j(q) = f_j [\exp(-a_j q^2) + b_j q^2 \exp(-c_j q^2)], \quad (9a)$$

$$f_i(q) = f_i \left[ 1 - \sum_{j=1}^5 f_j(q)/1 - \sum_{j=1}^5 f_j \right], \quad (9b)$$

$$E_i(q) = E_i + \frac{q^2}{2m} [1 - \exp(-\alpha q^2)], \quad (9c)$$

$$\gamma_{i,j}(q) = \gamma_{i,j} + \alpha q + \beta q^2. \quad (9d)$$

where the values of the various constants are given in [16]. It may be shown that the use of Eqs. 9(a)–9(d) for the extension of the Drude functions to  $q > 0$  satisfies the sum-rules Eqs. 6(a) and 6(b) for all values of  $q$  [16]. The form of  $f_j(q)$  has been deduced from gas-phase data on  $\text{H}_2\text{O}$ ;  $f_i(q)$  is then obtained by the application of the  $f$ -sum-rule. Eqs. 9(c) and 9(d), on the other hand, are based on the experimental Bethe ridge of liquid water. Specifically, the expression of  $E_i(q)$  has been deduced from an analysis of the observed line-peak of the Bethe ridge. For large  $q$ -values it reduces to the standard expression of the impulse approximation, i.e. a line-peak along the free-electron limit. For the discrete excitations we assume  $E_j(q) = E_j(0) \equiv E_j$  since the application of Eq. (9c) (or its impulse approximation) to the discrete spectrum would

Table 1

The Drude model parameters obtained in the present study for the amorphous (left column) and hexagonal (right column) forms of ice

| Transition                       | $E_{i,j}$ (eV) | $\gamma_{i,j}$ (eV) | $f_{i,j}$ |      |        |        |
|----------------------------------|----------------|---------------------|-----------|------|--------|--------|
| <i>Excitations (j)</i>           |                |                     |           |      |        |        |
| (A <sup>1</sup> B <sub>1</sub> ) | 8.65           | 8.65                | 1.6       | 1.6  | 0.0090 | 0.0168 |
| (B <sup>1</sup> A <sub>1</sub> ) | 10.50          | 10.50               | 2.5       | 1.5  | 0.0096 | 0.0065 |
| (Ryd A + B)                      | 12.60          | 12.60               | 3.5       | 3.0  | 0.0210 | 0.0190 |
| (Ryd C + D)                      | 14.10          | 14.10               | 3.0       | 2.7  | 0.0040 | 0.0110 |
| (Diffuse bands)                  | 14.50          | 14.50               | 2.5       | 1.5  | 0.0030 | 0.0044 |
| <i>Ionizations (i)</i>           |                |                     |           |      |        |        |
| (1b <sub>1</sub> )               | 15.40          | 15.80               | 5.7       | 4.6  | 0.1250 | 0.1000 |
| (3a <sub>1</sub> )               | 18.60          | 18.00               | 7.1       | 7.5  | 0.1300 | 0.2000 |
| (1b <sub>2</sub> )               | 24.50          | 24.50               | 15.0      | 14.0 | 0.1100 | 0.1100 |
| (2a <sub>1</sub> )               | 38.00          | 35.00               | 30.0      | 30.0 | 0.4110 | 0.3580 |
| (K-shell) <sup>a</sup>           | 450            | 450                 | 360       | 360  | 0.3143 | 0.3143 |

<sup>a</sup> Values based on oxygen's K-shell atomic-oscillator-strength (see text).

be unrealistic. The form of  $\gamma_{i,j}(q)$  has been deduced from the line-width of the ridge. Note that, to a first approximation and in the absence of more detailed (i.e. shell-specific) data, the parameters  $\alpha$ ,  $\beta$  in Eqs. 9(c) and 9(d) are assigned the same values for all transitions  $i, j$ . Likewise, although the dispersion of  $f_{i,j}(q)$  for condensed water (liquid or solid) will differ from that of the gas phase, Eqs. 9(a) and 9(b) are expected to provide a reasonable first approximation. Compared to earlier used models, the above dispersion scheme substantially improves the agreement with the experimentally measured Bethe ridge of liquid water [16,17]. In particular, the modified form of the impulse approximation (Eq. (9c)) and the dispersion of the damping coefficient (Eq. (9d)) correct previous models in terms of both the position and width of the Bethe ridge (especially at moderate values of  $q$ ), as also noted by others [28]. Finally, it is of practical interest to note that, despite the lack of theoretical rigor, the above model preserves the simple mathematical form of the extended-Drude model and it is, therefore, particularly convenient for further computations.

### 3. Results and discussion

Fig. 1 presents a comparison between our model calculations for all dielectric function representations, namely,  $\text{Im}(\epsilon)$ ,  $\text{Re}(\epsilon)$  and  $\text{Im}(-1/\epsilon)$ , and the available experimental data for the valence domain (panels a and b), as well as, the atomic calculations at higher energy transfers where the K-shell sets in (panel c). The very good agreement between our model with the data is evident and reveals the effectiveness of the present methodology. With respect to the 21 eV peak in the  $\text{Im}(-1/\epsilon)$  spectrum (panels a and b), the values for amorphous and hexagonal ice are  $\sim 0.65$  and  $\sim 0.75$ , respectively. These values are closer to what has been obtained for liquid water by the IXS measurements ( $\sim 0.65$ ) [29] than by the earlier reflectance measurements ( $\sim 1.1$ ) [11]. Likewise, the values of our models for  $\text{Re}(\epsilon)$  at 21 eV are significantly larger than zero and closer to the IXS data. As it has been already noted [30], both observations support the hypothesis that the 21 eV peak, although perhaps of some collective character, is not due to a plasmon excitation. Theory predicts that at sufficiently high energy transfers (so that  $\text{Re}(\epsilon) \approx 1$ ) the absorption spectrum of the gas and condensed phases become identical since any screening effects provided by the electronic subsystem of the condensed phase gradually vanish as the wavelength of the field decreases. Consistent with the above are the results of Fig. 1(c) which shows that our model for solid water is in almost excellent agreement above  $\sim 100$  eV with the atomic calculations for  $\text{H}_2\text{O}$  (relevant to the gas phase) of the NIST database. In the inset of Fig. 1(c) we present the asymptotic behavior of  $\text{Re}(\epsilon)$  as calculated by our model. At the energy range where the K-shell sets in ( $>540$  eV),  $\text{Re}(\epsilon)$  approaches unity to within 0.01% which is indicative of the system's atomic behavior.

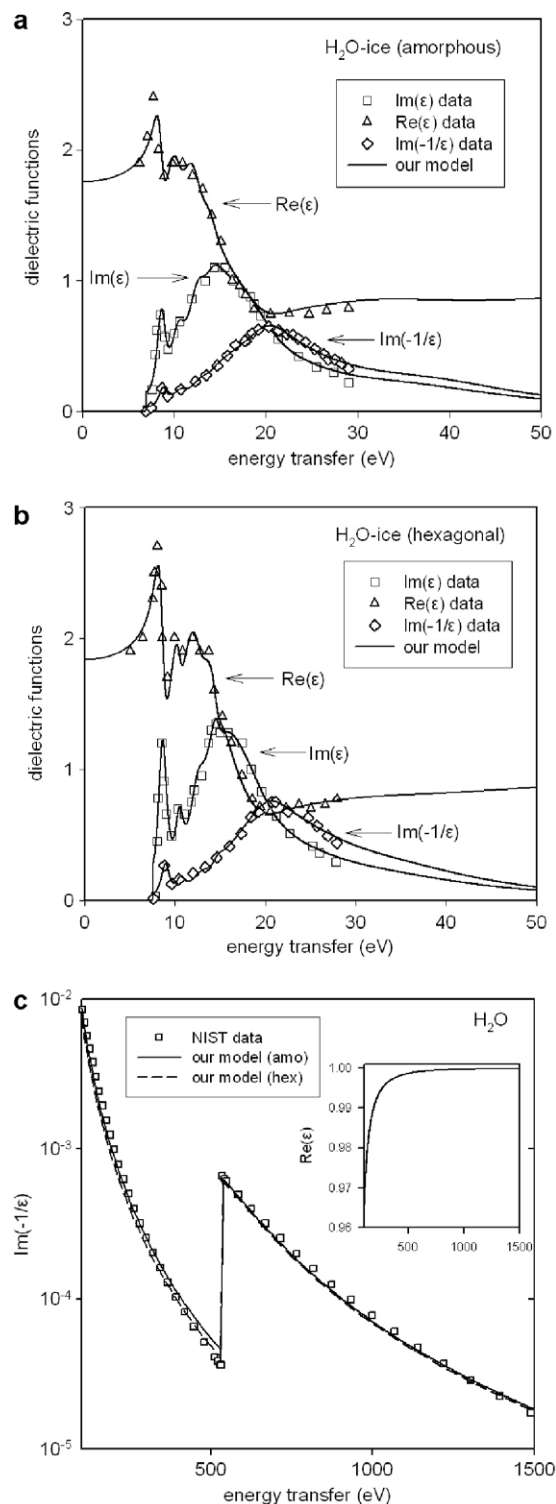


Fig. 1. A comparison of our optical-data models for the dielectric functions of the two forms of ice against the experimental data at the valence range (panels a and b) obtained by electron-energy-loss-spectroscopy [19] and reflectance measurements [20], and the atomic calculations of NIST [27] at the inner shell range (panel c).

In general, the typical differences observed between the gas and condensed phase absorption spectrum are attributed to intermolecular interactions in the condensed phase



which tend to smear, broaden, and attenuate the characteristic low energy peaks observed for single molecules. Similar to the liquid phase, the spectrum of ice is harder (blue-shifted) compared to the gas phase exhibiting a prominent peak at  $\sim 21$  eV. To further investigate the above effects, we present in Fig. 2 our preliminary time-dependent density-functional-theory (TDDFT) calculations of the optical oscillator strength spectra of a single  $\text{H}_2\text{O}$  molecule and a cluster of  $4\text{H}_2\text{O}$  molecules and compare with experimental data of water in the gaseous [31], liquid [29], and solid [19,20] phase. The TDDFT overcomes the well-known limitations of the (static) DFT for excited electronic states [32] and has very recently been applied with considerable success to the dielectric response of liquid water over most of the valence domain [8]. It is also computationally simpler than the Bethe–Salpeter approach based on the many-body Green’s function [33] recently applied to hexagonal ice [7]. Our TDDFT calculations were carried out with the aid of Gaussian 03 using the 6-31+g(dp) basis set [34], which includes polarization and diffuse functions for an adequate description of hydrogen-bonding. They were carried out at the optimum geometry in each case, single  $\text{H}_2\text{O}$  molecule or cluster of  $4\text{H}_2\text{O}$  molecules, obtained by geometry optimization with the aid of DFT-B3LYP [34] calculations employing the same basis set as above. Although of limited spectral range, they cover an important part of the valence band from threshold to  $\sim 20$  eV. The TDDFT results seem to support the above mentioned effects of the intermolecular forces.

The internal consistency of the dielectric functions is examined through the application of the  $f$ -sum-rule and the calculation of the  $I$ -value of Bethe’s stopping theory. The former defines the “effective” number of target elec-

trons ( $= Z \times N_{\text{eff}}$ , where  $Z = 10$  for water), that participate in the inelastic interaction whereas the latter an “effective” mean excitation energy for the target. In Fig. 3, we present the  $N_{\text{eff}}$  and the  $I$ -value as a function of the maximum energy transfer. The  $N_{\text{eff}}$  is partitioned to its K-shell and valence band contribution. The former asymptotically approaches the value 0.178 (consistent with accurate atomic calculations [26]), while the latter the value  $1 - 0.178 = 0.822$  (to within 1%) which is expected from theory. The calculated  $I$ -values for hexagonal and amorphous ice are 80.5 and 84.7 eV, respectively. These values are close to liquid water calculations by us (82.4 eV) [16] and Dingfelder (81.8 eV) [25] and to recent experiments by Bichsel ( $79.7 \pm 0.5$  eV) [35]. They are noticeable higher than the recommended ICRU value for liquid/solid water (75 eV) [36]. The variation of  $N_{\text{eff}}$  and  $I$  with the maximum energy transfer depicted in Fig. 3 has important implications for the applicability of Bethe’s stopping power formula. Over the range of energy transfers (and, thus, impact energies) where both  $N_{\text{eff}}$  and  $I$  remain constant

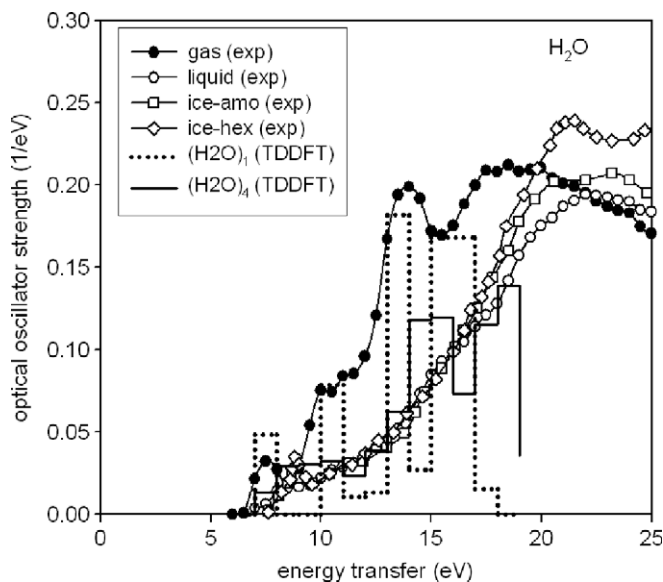


Fig. 2. A comparison of our TDDFT calculated optical-oscillator-strength of a single  $\text{H}_2\text{O}$  molecule and a cluster of  $4\text{H}_2\text{O}$  molecules against experimental data for various forms of water (gas [31], liquid [29], ice-amorphous [19], ice-hexagonal from [20]).

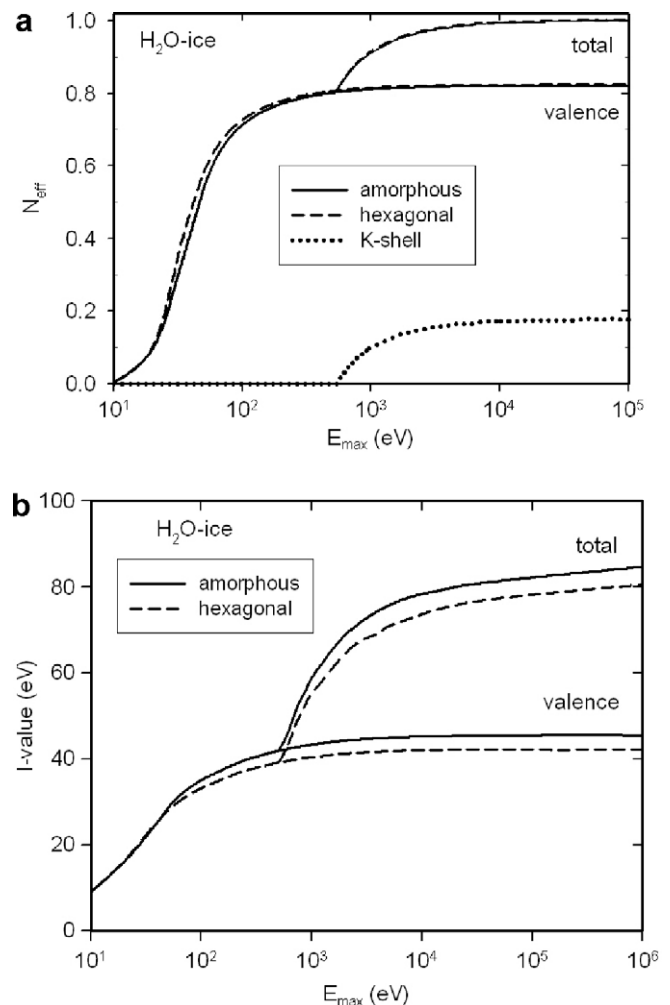


Fig. 3. The  $N_{\text{eff}}$  (panel a) and the  $I$ -value of stopping theory (panel b) as a function of the maximum energy transfer in an inelastic collision for the two forms of ice.

and equal to their asymptotic values, Bethe's formula is expected to be reasonably accurate whereas at lower energies corrections are needed to account for deviations of  $N_{\text{eff}}$  and  $I$  from their asymptotic values. For water ice, for example, the Bethe cut-off for electron impact is expected to be at the 1–10 keV range.

The dispersion characteristics of the (single-scattering) energy-loss-function (ELF)  $\text{Im}[-1/\epsilon(E, q)]$  are depicted in Figs. 4 and 5. Specifically, in Fig. 4 we present the energy distribution of the ELF for some fixed values of momentum transfer ( $q$ ), whereas in Fig. 5 we present its momentum distribution for some fixed values of energy transfer ( $E$ ). We may notice from Fig. 4 that, whereas at  $q = 0$  both the low energy sharp excitation peak and the broad absorption peak at  $\sim 21$  eV of hexagonal ice are noticeable higher than the ones of amorphous ice, the differences gradually vanish with increasing  $q$  and are almost indistinguishable at  $q = 2$  a.u. In general, with increasing  $q$  the ELF is shifted to higher energies and band-structure effects become gradually unimportant. In Fig. 5(a) we depict the momentum distribution at the valence loss range ( $E < 50$  eV), whereas in Fig. 5(b) the distributions are extended to higher energy transfers where the Bethe ridge becomes noticeable. For both forms of ice, the momentum distribution of the ELF acquires its characteristic Bethe ridge shape for energy transfers above  $\sim 4$  Ry. However, it is not till we reach  $\sim 10$ – $12$  Ry that the free-electron condition  $E(\text{Ry}) = q^2$  (a.u.) for the ELF maximum is obtained. In contrast to the sharply peaked Bethe ridge of earlier extended-Drude and  $\delta$ -oscillator models, the present dispersion scheme predicts a significantly broader momentum distribution for the ELF at the Bethe ridge which is consistent with both experimental observations [30,37,38] and more elaborate theoretical models [39,40].

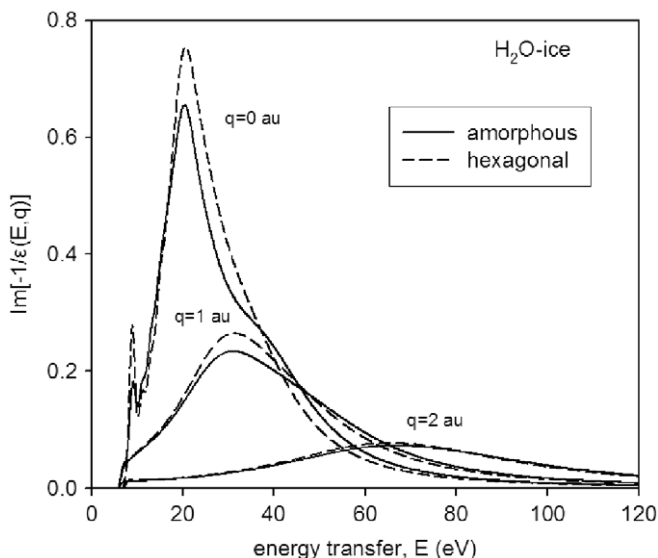


Fig. 4. The energy distribution of our model energy-loss-functions for the two forms of ice at different values of momentum transfer.

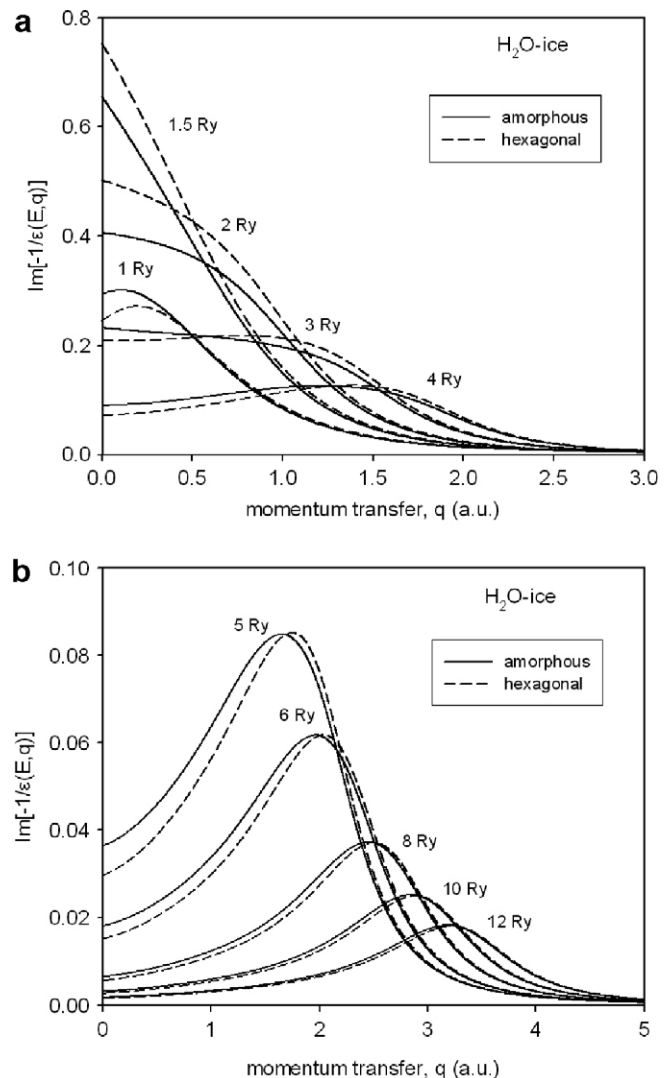


Fig. 5. The momentum distribution of our model energy-loss-functions for the two forms of ice at different values of energy transfer. Panel a emphasizes the valence band range whereas panel b the energy range associated with the Bethe ridge.

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